DOE ASPECT · DE-FOA-0003647 · TOPIC AREA 1, SUBTOPIC 1a

Bench-Scale Process Development for Enzymatic Production of Terephthalic Acid from Post-Consumer PET

PlastiBioFuel LLC, Haslet, Texas. Michael David Ward, Founder and CEO, michael@plastibiofuel.com, 817-319-7383. Federal share sought $2,000,000 over 24 months, with the minimum 20 percent cost share.

1. THE CHEMICAL, THE FEEDSTOCK, AND THE FIT

Objectives of the NOFO

The target chemical is terephthalic acid, the principal monomer of polyethylene terephthalate. PET is the most abundant polyester plastic, with almost 70 million tons manufactured annually worldwide (Tournier et al., Nature 2020, DOI 10.1038/s41586-020-2149-4), which places terephthalic acid an order of magnitude and more above the one million metric ton global market threshold this NOFO sets. It is a chemical product, not a fuel, and it is not ethanol.

The feedstock is sorted municipal solid waste, specifically its plastic constituent. Appendix A lists sorted municipal solid waste as an acceptable feedstock and defines it as the organic and plastic constituents of the municipal solid waste stream going to the landfill. PET is that plastic constituent. The project proposes no feedstock outside the Appendix A table, which is exclusive: an application proposing a feedstock the table does not identify will not be considered. The material is domestic, abundant, priced well below agricultural feedstock, and does not compete with food production. Producing terephthalic acid from it returns a high-volume monomer to domestic polymer manufacture from material that is currently landfilled, which strengthens a domestic supply chain and displaces imported monomer.

2. TECHNOLOGY, STATE OF THE ART, AND WHAT IS NEW

Uniqueness and innovation

Enzymatic PET depolymerization is limited by reactor engineering rather than by enzyme discovery. FAST-PETase, the engineered hydrolase this process uses, was reported by the Alper group at the University of Texas at Austin in Nature in 2022 and was itself produced by machine learning. PlastiBioFuel did not invent the enzyme and makes no claim to it.

PlastiBioFuel's contributions, covered by U.S. provisional patent application 63/848,456, are the immobilization of that enzyme in a TAPB-BPDA covalent organic framework, a continuous spiral-flow

reactor geometry derived computationally, and the integrated process that connects them. The reactor geometry exists as a validated simulation model and not as hardware; no reactor wet-body material has been qualified, and the candidate resin class remains a screening candidate pending soak-coupon qualification as of 2026-10-01. Nothing in this concept paper depends on hardware that does not exist.

The advance this project proposes is to resolve a transport limitation that the preliminary data has already located, and to convert the simulation model into a bench-validated design tool with quantified error. The company runs a wet-laboratory program and a simulation and modeling program that feed each other, and the principal built and runs the modeling himself.

3. PRELIMINARY DATA

Feasibility

The authoritative source is the University of North Texas Final Project Report for Phase I, signed by Prof. Shengqian Ma and dated 2026-08-07. Standard conditions throughout: pH 8, 100 mM phosphate buffer, 1 mL, 100 mg PET, 0.05 percent Tween-20, 65 °C.

DEPOLYMERIZATION CONFIRMED
Free FAST-PETase breaks real post-consumer water bottle PET down to terephthalic acid, confirmed by HPLC against standards.

IMMOBILIZATION CONFIRMED
The enzyme loads into the TAPB-BPDA covalent organic framework with no protein detectable in the supernatant at 280 nm.

IMMOBILIZED TURNOVER
Measured at 122 µmol/h/mg at 24 hours, against a 246 to 348 literature range for free enzyme.

CAUSE OF THE GAP
The laboratory attributes the gap to mass transfer down the framework channel plus insufficient PET pre-treatment.

SCHEDULE
Three milestones closed ahead of schedule against a planned 2026-06-30 date.

A separate run at 80 °C gave roughly 3.6 times the standard rate. That figure is confounded because water evaporated at that temperature, and the laboratory reads the gain as improved substrate access rather than enzyme kinetics. It is reported here with that caveat and is not relied on as a rate measurement.

WHY THIS IS A TRANSPORT PROBLEM

Two independent observations converge on one cause. The laboratory attributes the gap between 122 and the 246 to 348 literature range to mass transfer down the framework channel. The temperature run is independently read as substrate access rather than kinetics. Transport is what pre-treatment, particle configuration, and reactor geometry decide, and none of those can be settled at 1 mL scale.

4. PROPOSED WORK

Subtopic Area 1a, two budget periods

Subtopic 1a is the correct entry point. The technology is past proof of concept and needs bench-scale process development and optimization. It is not ready for the integrated pre-pilot system

at half a dry ton per day that Topic Area 2 requires, and the work does not involve strain discovery, powder-form or heterogeneous chemical catalysis, thermocatalytic processing, or electrocatalytic cells.

Budget Period 1, verification	Reproduce the immobilized turnover measurement under controlled evaporation, and separate transport from kinetics by varying pre-treatment and framework particle configuration at fixed enzyme loading. Characterize the PET pre-treatment state that the report identifies as insufficient. Closes on a go or no-go decision point.
Budget Period 2, process development	Optimize pre-treatment, enzyme loading, and framework configuration against measured turnover; establish an operating envelope with quantified uncertainty against the simulation model; and recover terephthalic acid to a purity specification acceptable for re-entry into polymer manufacture.

The PET fraction of sorted municipal solid waste is the real feedstock throughout. Model compounds are permitted under Subtopic 1a through Budget Period 1 only, and this project does not require them.

5. RISKS, CHALLENGES, AND MITIGATION

Technical, regulatory, financial

TRANSPORT MAY NOT FULLY RELIEVE

The limitation may prove only partly addressable at bench scale. The Budget Period 1 go or no-go is designed to answer exactly that question before Budget Period 2 funds are committed.

PRE-TREATMENT ENERGY COST

Aggressive pre-treatment could erode process economics. Energy and mass balance for each pre-treatment state is measured in Budget Period 2 as part of the operating envelope rather than assumed.

PRODUCT PURITY

Terephthalic acid is an existing commodity chemical, so no new substance registration is required. Recovered monomer re-entering polymer manufacture must meet a purity specification, which is a Budget Period 2 measurement. Food-contact end uses would require the converter's own clearance, not PlastiBioFuel's.

INTELLECTUAL PROPERTY

Provisional application 63/848,456 was filed in July 2025 and its conversion is being settled with patent counsel. FAST-PETase is published prior art from UT Austin, and the company's freedom-to-operate position on it is under review with the same counsel. Both are diligence items, not technical obstacles.

SUBAWARD TERM

Sponsored Research Agreement SRA1016 with the University of North Texas runs to 2027-08-31, before a 2027 federal period of performance would end. An amendment extending the term is required and is in progress.

COST SHARE

PlastiBioFuel will provide the required minimum 20 percent cost share. Sources are documented at Stage 1, where the NOFO requires them.

6. QUALIFICATIONS AND RESOURCES

Capability to perform

PlastiBioFuel LLC is a service-disabled veteran-owned small business, certified and active in SAM.gov, UEI KQAWZ54RDUM8, CAGE 175D6, NAICS 325199, 325193, and 541715. Michael David Ward, Founder and CEO, is the principal investigator and built and runs the simulation and

modeling program the design rests on.

Wet-laboratory work is performed at the University of North Texas under Sponsored Research Agreement SRA1016. Prof. Shengqian Ma leads framework and immobilization chemistry and has a published record in enzyme immobilization in covalent organic and metal-organic frameworks; he participates as other personnel on the subaward rather than as senior or key personnel of the prime. Higher education is eligible as a subrecipient under Topic Area 1.

7. IMPACT OF DOE FUNDING

Impact on the field and the application

Without this award the transport question stays unresolved at 1 mL scale, and the simulation model stays unvalidated against bench data. With it, the project produces a quantified operating envelope for enzymatic PET depolymerization and a bench-validated route from landfill-bound plastic to polymer-grade terephthalic acid. That is the result the field needs in order to size a pre-pilot system under Topic Area 2, and it is the result that would let PlastiBioFuel return to this program with prior scale data in hand.

8. OTHER FEDERAL APPLICATIONS PENDING

Disclosure

PlastiBioFuel LLC has one other federal proposal pending: NSF SBIR Phase I, Research.gov proposal 328095, solicitation NSF 26-510, subtopic BT10, total requested $304,955, against the November 4, 2026 window. That proposal covers covalent organic framework and enzyme reusability, the monomer-to-ethanol biological pathway, and construction of an integrated lab-scale spiral reactor module. The ASPECT scope proposed here covers none of those three. There are no shared tasks, no shared budget lines, and no dependency in either direction.

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